
Report

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Abstract

In recent years, semi-supervised learning has emerged as a powerful paradigm for leveraging unlabeled data to improve model performance, especially in scenarios where labeled data is scarce. This study explores the application of three semi-supervised learning algorithms—Safe Student, DeFixMatch, and ReFixMatch—across different data modalities: images, audio, and text. We apply Safe Student to the STL-10 image dataset, DeFixMatch to the Speech Commands audio dataset, and ReFixMatch to the AG News text dataset. Through extensive experiments involving exploratory data analysis, data augmentation, and tailored model architectures, we demonstrate the effectiveness of these algorithms in enhancing classification performance. The models achieved test accuracies of 65.8% on STL-10, 92.28% on Speech Commands, and 92.83% on AG News. Our findings highlight the potential of semi-supervised learning methods in diverse domains and provide insights into the challenges and considerations specific to each data type.

Semi-Supervised Learning, Safe Student, DeFixMatch, ReFixMatch, Data Augmentation, Image Classification, Audio Classification, Text Classification

1 Introduction

1.1 Background

The rapid growth of data generation in various fields has led to an abundance of unlabeled data, while acquiring labeled data remains costly and time-consuming. Semi-supervised learning (SSL) addresses this challenge by utilizing both labeled and unlabeled data to improve model performance. SSL is particularly valuable in scenarios where labeled data is scarce but unlabeled data is plentiful, such as in image recognition, speech processing, and natural language understanding.

In image classification, models often require large amounts of labeled images to achieve high accuracy. Similarly, in speech recognition and text classification, collecting and annotating data can be labor-intensive. SSL methods aim to leverage the structure in unlabeled data to learn better representations and decision boundaries.

Recent advancements in SSL, such as Safe Student, DeFixMatch, and ReFixMatch, have shown promising results by combining pseudo-labeling with consistency regularization and data augmentation. These methods reduce reliance on labeled data while maintaining or even improving model performance.

1.2 Literature Survey / Related Work

The Safe Student algorithm enhances SSL by mitigating the impact of incorrect pseudo-labels through a safe training mechanism. It dynamically adjusts the pseudo-labeling process to reduce the risk of propagating errors, making it suitable for datasets where incorrect pseudo-labels can significantly degrade performance.

DeFixMatch builds upon FixMatch by incorporating distribution alignment and debiasing techniques. It addresses the confirmation bias in pseudo-labeling by adjusting class distributions, leading to improved performance in imbalanced datasets and enhancing generalization.

ReFixMatch further refines the FixMatch framework by introducing adjustments to the confidence thresholding and augmentation policies. It aims to reduce the negative impact of incorrect pseudo-labels and improve robustness across different data domains.

Previous studies have applied these algorithms primarily to image datasets. However, there is limited exploration of their effectiveness in other modalities such as audio and text. Our work extends the application of these SSL methods to the Speech Commands audio dataset and the AG News text dataset, providing a comparative analysis across different domains.

1.3 Objectives

The primary objectives of this study are:

- **Apply Semi-Supervised Learning Algorithms:** Implement Safe Student, DeFixMatch, and ReFixMatch on image, audio, and text datasets, respectively, to evaluate their effectiveness in different data modalities.
- **Perform Exploratory Data Analysis and Data Augmentation:** Conduct thorough EDA on each dataset to understand their characteristics and apply appropriate data augmentation techniques tailored to each modality.
- **Analyze and Compare Performance:** Assess the performance of each model using relevant evaluation metrics and analyze the results to identify strengths and weaknesses.
- **Provide Insights and Recommendations:** Offer insights into the applicability of each SSL method across different domains and recommend best practices for future work.

1.4 Scope

This study focuses on three datasets representing different data types:

- **STL-10 Dataset (Images):** An image recognition dataset used to assess the effectiveness of the Safe Student algorithm in image classification tasks with limited labeled data.
- **Speech Commands Dataset (Audio):** An audio dataset of spoken words used to evaluate DeFixMatch in the context of audio classification and speech recognition.
- **AG News Dataset (Text):** A text dataset comprising news articles used to test the applicability of ReFixMatch in text classification tasks.

We limit our experiments to one SSL algorithm per dataset, chosen based on their strengths and suitability to the data type. The study also involves data preprocessing, augmentation, and the use of specific model architectures appropriate for each modality.

1.5 Impact

By extending the application of SSL algorithms beyond image data to audio and text domains, this work contributes to the broader understanding of semi-supervised learning’s versatility. The insights gained from this study can aid practitioners in selecting suitable SSL methods for various data types and inform future research in developing more robust and generalizable algorithms.

The successful application of these methods has the potential to reduce the dependency on large labeled datasets, lowering costs and accelerating development in fields such as computer vision, speech recognition, and natural language processing. Additionally, our comparative analysis provides valuable information on the strengths and limitations of each algorithm, guiding their adaptation and improvement for diverse applications.

2 Materials and Methods

2.1 Datasets

2.1.1 STL-10 Dataset

The **STL-10 dataset** contains

$$96 \times 96$$

color images belonging to 10 classes. It includes 5,000 labeled images for training, 8,000 labeled images for testing, and 100,000 unlabeled images ideal for semi-supervised learning experiments.

2.1.2 Speech Commands Dataset

The **Speech Commands dataset** comprises one-second audio recordings of 35 spoken words, with approximately 85,000 audio files for training and 11,000 for testing. This dataset facilitates exploration of semi-supervised learning in audio classification.

2.1.3 AG News Dataset

The **AG News dataset** is a text classification benchmark with news articles categorized into four classes, totaling 120,000 articles for training and 7,600 for testing. It's suitable for evaluating semi-supervised learning in natural language processing.

2.2 Methodology

2.2.1 Data Preprocessing

STL-10 Dataset: Images were resized to 64 x 64 pixels and normalized per channel. We used 1,000 labeled images to simulate limited labeled data, with 20% set aside for validation, and utilized the unlabeled set for semi-supervised learning.

Speech Commands Dataset: Audio files were loaded at 16 kHz. We generated Mel Spectrograms using 64 filters and converted them to decibel units. The predefined training and validation splits were used, limiting labeled data to simulate semi-supervised learning.

AG News Dataset: Text was converted to lowercase, tokenized using the pre-trained model's tokenizer, and sequences were padded or truncated to a maximum of 128 tokens. The data was split into 90% training and 10% validation sets, with a portion treated as unlabeled.

2.2.2 Data Augmentation Techniques

STL-10 Dataset: Applied random horizontal flipping (50% probability), random cropping resized to 64 x 64, and color jittering (adjusting brightness, contrast, saturation, and hue) to increase data diversity.

Speech Commands Dataset: Employed time masking and frequency masking to spectrograms, and applied time stretching and pitch shifting to mimic variations in speech patterns, enhancing model robustness.

AG News Dataset: Used NLTK's WordNet for synonym replacement, and introduced variability through random insertion and deletion of words, aiming to improve robustness while preserving semantic meaning.

2.2.3 Model Architectures

STL-10 Dataset: Utilized a pre-trained *ResNet-18* model, modifying it by replacing the final layer to output 10 classes and initializing new layer weights using Xavier initialization.

Speech Commands Dataset: Developed a custom CNN with convolutional layers and ReLU activations, incorporating batch normalization and max-pooling layers. Adaptive average pooling handled variable input sizes, followed by fully connected layers with softmax output.

AG News Dataset: Employed a pre-trained *BERT-Tiny* model, adding a classification head for four classes and re-initializing classification layer parameters. This leveraged transfer learning and reduced computational requirements.

2.3 Novelty

Our work extends algorithms to audio and text data, demonstrating their versatility across modalities. We developed domain-specific augmentation techniques and employed adaptive pooling layers to handle variable input sizes. Additionally, we integrated NLTK’s WordNet for effective synonym-based text augmentations.

2.4 Evaluation Metrics

We used **accuracy** to measure the proportion of correct predictions. **Precision**, **recall**, and **F1-score** provided insights into model performance, with precision as true positives over predicted positives, recall as true positives over actual positives, and F1-score as their harmonic mean. Confusion matrices visualized actual versus predicted classifications, aiding error analysis.

3 Results

3.1 Exploratory Data Analysis

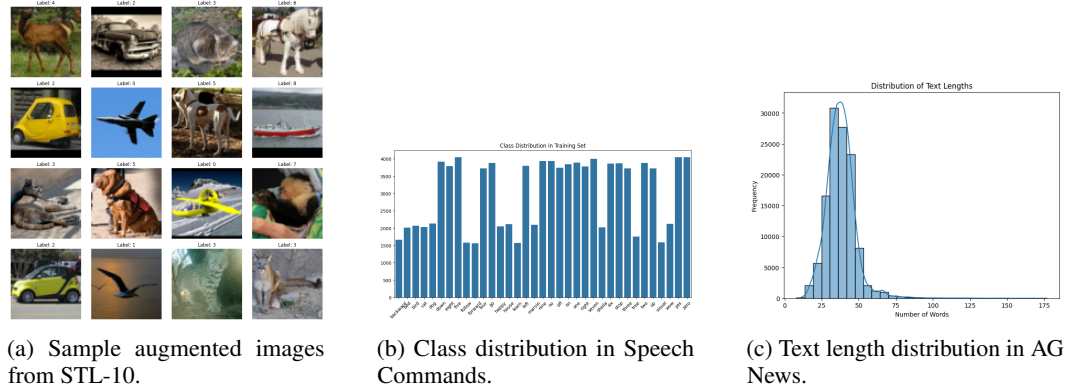


Figure 1: Exploratory Data Analysis of the datasets.

3.2 Model Performance

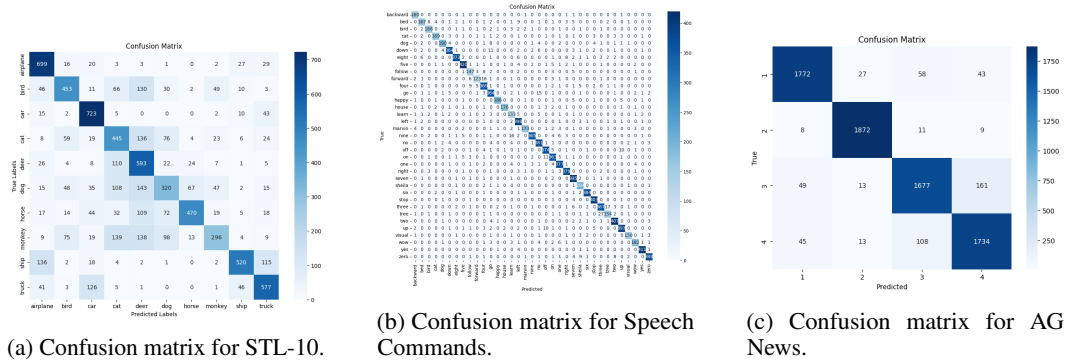


Figure 2: Confusion matrices for the test sets.

4 Discussion

In this study, we applied semi-supervised learning algorithms to image, audio, and text data, each presenting unique challenges.

4.1 STL-10 with Safe Student

Applying **Safe Student** to the STL-10 dataset yielded a test accuracy of **65.8%**. While it leveraged unlabeled data, the accuracy indicates room for improvement due to:

- **Complex Image Data:** High variability in STL-10 images may have limited effectiveness.
- **Pseudo-Label Quality:** Incorrect pseudo-labels negatively impacted learning.

4.2 Speech Commands with DeFixMatch

Using **DeFixMatch** on the Speech Commands dataset achieved a test accuracy of **92.28%**. Key factors include:

- **Effective Use of Unlabeled Data:** Consistency regularization improved performance.
- **Audio Augmentations:** Time and frequency masking enhanced model robustness.

4.3 AG News with ReFixMatch

Employing **ReFixMatch** on the AG News dataset led to a test accuracy of **92.83%**, demonstrating:

- **Adaptability to Text Data:** Successful application in text classification.
- **Enhanced Data Diversity:** Text augmentations like synonym replacement were effective.

4.4 Comparative Analysis

The results highlight the importance of aligning algorithms with data types. DeFixMatch and ReFixMatch performed well on audio and text data, respectively, while Safe Student may need optimization for image data.

4.5 Challenges

- **Hyperparameter Tuning:** Careful adjustment was essential for each algorithm.
- **Computational Limitations:** Resource constraints limited extensive experimentation.

5 Work to be Done

Future work includes:

- **Cross-Appling Algorithms:** Test DeFixMatch on STL-10 and Safe Student on AG News and Speech Commands.
- **Hyperparameter Optimization:** Conduct thorough searches for optimal settings.
- **Enhanced Augmentations:** Explore additional techniques specific to each modality.
- **Algorithm Comparison:** Compare the algorithms on a common dataset.
- **Ensemble Methods:** Combine algorithms to leverage complementary strengths.

6 Distribution of Work Among Group Members

Kanak (Roll Number: 2022232)

Kanak was responsible for exploratory data analysis, implementing data augmentations, conducting testing, and compiling the report.

Ben Jamir (Roll Number: 2022130)

Ben focused on implementing the semi-supervised learning algorithms (SSL and OSSL), benchmarking performance, analyzing outputs, and preparing the presentation slides.

References

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